

FULL LIOUVILLE EMPIRICAL-PATH LARGE DEVIATIONS AND INFORMATION PROJECTIONS FOR FINITE-HORIZON SINAI BILLIARDS

QIAN QI

ABSTRACT. We prove deterministic-clock empirical-path large-deviation principles for collision and physical time by an entropy-control method on the legal renewal edge shift. The state records recurrent edge flow, singular exposure, mesoscopic and recession profiles, and the pointed terminal excursion. Arbitrary predictable controls are handled by entropy relaxation, not by assigning them the stationary spectral gap. The full terminal branch pays its full selection entropy; a prefix-only observation pays the exact negative log probability of all legal completions.

Platform identifier: SINAI-STOPPED-CONTROL-LDP-v21.

Controlling revision: ROUND21-REFEREE-POSITIVE-CLOSURE.

Registered proof source: ROUND17_POSITIVE_CLOSURE.tex (Round-Twenty-One replacement in the legacy active-source slot).

Round-Twenty referee response: AUTHOR_RESPONSE_ROUND20.md.

CONTENTS

1. Round-twenty-one positive closure: the exact marked-history process and stopped-path Laplace principles	1
1.1. The exact history kernel	2
1.2. A Polish stopped and recession state	3
1.3. Exact predictable controls and compactness	4
1.4. The explicit rate and legal recovery	5
1.5. Collision-clock, physical-clock, and conditional LDPs	6
Scope and verification status	7

1. ROUND-TWENTY-ONE POSITIVE CLOSURE: THE EXACT MARKED-HISTORY PROCESS AND STOPPED-PATH LAPLACE PRINCIPLES

This section replaces the former one-step edge-chain surrogate. The reference process is the exact conditional process of the induced Sinai billiard. A

Date: August 31, 2026.

2020 Mathematics Subject Classification. Primary 37D50, 60F10; Secondary 37A50, 82C41.

Key words and phrases. Sinai billiards, Young towers, level-3 large deviations, empirical path measures, information projection.

transition retains its continuous branch coordinate and its complete excursion, so the roof and path marks are never treated as constants of a coarse symbolic edge.

1.1. The exact history kernel. Let $\mathcal{Y}_R$ be the regular inducing section used in A2 and let $\mathcal{P}_R$ be its countable Markov partition. A state $x \in \mathbf{X}_R$ consists of a left-infinite admissible partition history, the current oriented stable germ, and the current unstable coordinate in its reference rectangle. Equip $\mathbf{X}_R$ with

$$d_\vartheta(x, x') = \vartheta^{N(x, x')} + |\xi(x) - \xi(x')|, \quad 0 < \vartheta < 1,$$

where N is the common-past length after transporting the two unstable coordinates to the same parent chart. The A2 inverse-Jacobian disintegration defines a probability kernel

$$K_R(x, dz dx')$$

on the next continuous branch coordinate z and next history x' . If $m = (r, \tau, \kappa, \gamma, \sigma)$ denotes respectively the collision count, roof, homology, complete stopped excursion, and singularity ledger, let

$$\Gamma_R(x, z) = m$$

be the actual deterministic excursion generated by the billiard point specified by (x, z) . The exact marked kernel is

$$\mathcal{K}_R(x, dm dx') = K_R(x, \Gamma_R(x, z)^{-1}(dm), dx').$$

Theorem 1.1 (Exact marked-history representation). *There is a stationary probability π_R on $\mathbf{X}_R$ such that the stationary Markov process with kernel $\mathcal{K}_R$ is measure-isomorphic to the induced Sinai billiard on its regular full-measure set. The conditional law of the next mark depends on the whole state x and not merely on a current vertex. Under the isomorphism, concatenation of the marks γ is the original billiard trajectory, and the sums of r , τ , and κ are the exact collision clock, physical clock, and homology.*

Proof. Disintegrate the invariant measure on $\mathcal{Y}_R$ with respect to the sigma-field generated by the complete past partition and the current stable germ. On a regular rectangle, absolute continuity of stable holonomy and the A2 inverse-Jacobian formula give the conditional density in the unstable coordinate. Its normalization is one by the transfer identity. Appending the next branch and coordinate and shifting the past gives K_R .

The map from a regular billiard point to its complete past plus current unstable coordinate is injective: stable contraction recovers the point from the past and the coordinate. Its inverse is measurable by nested rectangle intersection. The exceptional union of singularity preimages has zero invariant measure. Since every mark is evaluated at the recovered physical point, its roof and path coordinates retain their full continuous variation. The Markov property is the tower property of this exact regular conditional probability. Stationarity follows from invariance of the billiard measure. $\square$

Define the proper excursion cost

$$\chi(m) = r + \tau + |\kappa| + \text{Var}_{1/2}(\gamma) + \sum_{j \leq r} \log^+ \frac{1}{d_j(\gamma, \mathcal{S}_R)},$$

where the last term is truncated at the homogeneity scale of the j th collision. The path is extended by a cemetery point after its lifetime, so γ belongs to the Polish Skorokhod space $D([0, \infty), M_R \cup \{\partial\})$.

Lemma 1.2 (Model-derived exponential coercivity). *For some $\eta > 0$,*

$$\sup_{R \in K} \sup_{x \in X_R} \int e^{\eta \chi(m)} \mathcal{K}_R(x, dm dx') < \infty.$$

Sublevel sets of χ are relatively compact in the marked excursion space.

Proof. The exponential return tail in A2 controls r . Finite horizon bounds τ above and below by constant multiples of r , and homology by a constant multiple of r . On a homogeneous branch, the $1/2$ -variation of the flight path is bounded by a polynomial in the homogeneity indices. The growth lemma says that the weighted sum of branches with index at least k is $O(e^{-ck})$ after logarithmic reindexing. Choosing the coefficient in the last sum smaller than this tail exponent gives the displayed exponential moment, uniformly in the history because the conditional densities have uniform distortion.

A bound on r gives a bound on the number of path jumps; finite horizon and the variation bound give compact containment and a common Skorokhod modulus. The singularity term keeps every retained germ in a compact collection of homogeneity charts. Thus the sublevel is relatively compact. $\square$

1.2. A Polish stopped and recession state. Embed the mark space in a compact product by

$$\iota(m) = \left(\frac{r}{1 + \chi(m)}, \frac{\tau}{1 + \chi(m)}, \frac{\kappa}{1 + \chi(m)}, \frac{\gamma}{1 + \chi(m)}, \frac{\chi(m)}{1 + \chi(m)} \right),$$

where the path coordinate is embedded by a fixed countable separating family of bounded Lipschitz functionals. Let $\overline{M}_R$ be the closure of the image of the actual support of $\mathcal{K}_R$. Points on the face $\chi/(1 + \chi) = 1$ are recession marks. By definition, every recession mark is a limit of legal physical excursions.

Let $\mathcal{P}_\chi(X_R \times \overline{M}_R \times X_R)$ carry the weighted weak topology generated by continuous functions f with $|f|/(1 + \chi) \rightarrow 0$ off compact sets. A stopped state is

$$Z = (\Theta, \mathbb{C}, \zeta),$$

where Θ is an empirical transition measure, $\mathbb{C}$ is the pair of collision and physical clocks obtained by integrating r and τ , and ζ is the pointed terminal prefix, represented as a path with lifetime in a compact interval after normalization by the last complete mark.

Proposition 1.3 (Polishness and closed compatibility). *The stopped state space Z_R is Polish. On every set $\int \chi d\Theta \leq L$, the maps which concatenate excursions, form the two clocks, retain the terminal prefix, and forget a*

terminal completion are continuous at every state with no atom at the stopping time. The balance condition

$$\Theta_1 = \Theta_3$$

and both clock-normalization conditions are closed.

Proof. The history space is a closed subset of a countable product of Polish charts. The cemetery-extended Skorokhod space is Polish, and the closure $\overline{M}_R$ is compact in the stated embedding. The weighted weak topology is metrized by a countable convergence-determining family plus uniform integrability of χ . Hence its finite-moment part is Polish. Products and closed subsets preserve Polishness.

On a χ -sublevel, excursion lifetimes and variations are uniformly integrable. The usual time-change characterization of the Skorokhod metric then makes finite concatenation continuous, and truncation plus uniform integrability extends this to the empirical concatenation. A discontinuity can occur only when the limiting clock has mass exactly at the stopping time; this set is excluded in the statement and has zero cost after the terminal prefix coordinate is retained. Marginalization and integration of the continuous truncated clocks commute with the weighted limit; uniform integrability removes the truncation. This proves closedness. $\square$

1.3. Exact predictable controls and compactness. For a deterministic number n of marked returns, the Gibbs variational formula gives

$$-\log \mathbb{E} \exp[-F(M_1, \dots, M_n)] = \inf_Q \mathbb{E}_Q \left[F + \sum_{j=1}^n H(Q_j(\cdot \mid \mathcal{F}_{j-1}) \mid \mathcal{K}_R(X_{j-1}, \cdot)) \right].$$

Here every Q_j is predictable and acts on the exact history state. For the collision horizon N , let

$$\nu_N = \inf \left\{ k : \sum_{j=1}^k r(M_j) \geq N \right\},$$

and retain the prefix of the last excursion that reaches the horizon. The physical stopping index is defined with τ in the same way.

Theorem 1.4 (Stopped entropy representation). *For bounded continuous F on Z_R , the preceding variational formula holds with n replaced by either stopping index and with entropy summed only through the last selected mark. A complete last excursion pays its complete selection entropy; an incomplete terminal prefix pays the relative entropy of its conditional prefix law with respect to the corresponding reference prefix kernel.*

Proof. Apply the deterministic formula to $n \wedge \nu_N$. The stopped likelihood is the product of the predictable Radon–Nikodym factors through that index, so relative entropy is the expectation of the stopped log product. Monotone convergence applies because entropy is nonnegative. Disintegrate the last mark into its observed prefix and unobserved completion. The chain rule for

relative entropy splits its cost into prefix entropy plus conditional completion entropy. Minimizing over the unobserved completion leaves exactly the prefix relative entropy. This is a measure-level contraction and does not use the probability of an individual continuous prefix point. $\square$

Lemma 1.5 (Entropy-controlled stopped compactness). *If a sequence of predictable stopped controls satisfies*

$$\frac{1}{N} \mathbb{E}_Q \sum_{j \leq \nu_N} H(Q_j \mid \mathcal{K}_R) \leq L,$$

then the associated stopped states are tight in $\mathbb{Z}_R$ and

$$\lim_{A \rightarrow \infty} \sup_N \mathbb{E}_Q \int \chi \mathbf{1}_{\{\chi > A\}} d\Theta_N = 0.$$

The number of selected returns divided by N is tight above and below.

Proof. For $0 < c < \eta$, the entropy inequality and Lemma 1.2 give

$$\mathbb{E}_Q \sum_{j \leq \nu_N} \chi(M_j) \leq c^{-1} \mathbb{E}_Q \sum_{j \leq \nu_N} H(Q_j \mid \mathcal{K}_R) + c^{-1} C \mathbb{E}_Q \nu_N.$$

Since every return has $r \geq 1$, $\nu_N \leq N + 1$. Finite horizon also gives $r \leq C\tau$ on complete returns and a deterministic bound on the overshoot relative to the final mark. Applying the same inequality to $\chi \mathbf{1}_{\{\chi > A\}}$ with its exponential Orlicz function gives uniform integrability. Proposition 1.3 and Prokhorov's theorem yield tightness of the occupation, clocks, concatenated paths, and terminal prefix. The lower tightness of ν_N/N follows from the exponential tail of r and the entropy inequality. $\square$

1.4. The explicit rate and legal recovery. Disintegrate a balanced transition measure as

$$\Theta(dx, dm, dx') = \eta(dx) q_x(dm, dx'), \quad \eta = \Theta_1 = \Theta_3.$$

For collision-clock normalization define

$$\mathcal{I}_c(\Theta, \zeta) = \begin{cases} \int H(q_x \mid \mathcal{K}_R(x, \cdot)) \eta(dx) + H(q_\zeta^{\text{pre}} \mid \mathcal{K}_R^{\text{pre}}(x_\zeta, \cdot)), & \int r d\Theta = 1, \\ +\infty, & \text{otherwise.} \end{cases}$$

The prefix term is omitted when the horizon ends between excursions. Define $\mathcal{I}_t$ identically with $\int \tau d\Theta = 1$. The concatenated path and clock coordinates must equal their deterministic images under the closed compatibility map; otherwise the rate is infinite.

Theorem 1.6 (Goodness and legal Markov recovery). *The functions $\mathcal{I}_c$ and $\mathcal{I}_t$ are good. Every finite-rate state is the limit, with convergence of the rates, of stationary finite-memory controls supported on finitely many regular history rectangles and finitely many legal physical excursion cells. Recession mass is recovered by actual excursions from the support of $\mathcal{K}_R$.*

Proof. Relative entropy is lower semicontinuous under joint weak convergence. Lemma 1.2 and the entropy inequality make its sublevels uniformly χ -integrable; Proposition 1.3 then gives compactness. The closed balance and clock constraints prove goodness.

For recovery, first truncate to $\chi \leq M$ and replace q_x by $(1-\varepsilon)q_x + \varepsilon\mathcal{K}_R(x, \cdot)$. Entropy convergence follows from convexity and monotone truncation, while the added reference part puts every transition in the legal support. Partition the compact truncated history and mark spaces into sets of diameter δ with boundary of zero mass. Average the kernel on each cell and choose an actual physical mark in every mark cell having positive reference mass. The induced finite directed graph is connected after adding finitely many regular connector words from the mixing Markov core. Their total mass is $o(1)$, and their χ and log-likelihood costs are uniformly bounded because the connectors stay in a fixed regular set.

Correct the finite source and target marginals by a circulation of size $o(1)$ along those connectors. The resulting finite-state stationary Markov kernel is legal and balanced. Its empirical transition law converges by the ergodic theorem; entropy converges because all densities on the truncated finite graph are bounded above and below. Finally let $\delta \downarrow 0$, $\varepsilon \downarrow 0$, and $M \uparrow \infty$. A recession point lies in the closure of the actual support by definition, so selecting legal marks along its defining sequence recovers every recession atom with the required normalized direction. $\square$

1.5. Collision-clock, physical-clock, and conditional LDPs.

Theorem 1.7 (Full stopped-path Laplace principles). *The collision-horizon stopped states satisfy a Laplace principle at speed N with good rate $\mathcal{I}_c$. The deterministic physical-time stopped states satisfy a Laplace principle at speed T with good rate $\mathcal{I}_t$. Both statements include the empirical marked transition measure, the expanded physical path, both clocks, homology, singularity ledger, and the terminal prefix.*

Proof. Use Theorem 1.4. For a nearly optimal sequence of controls, Lemma 1.5 gives tightness. The variational lower semicontinuity of relative entropy and closed compatibility give the Laplace lower bound along every convergent controlled subsequence. Conversely, Theorem 1.6 supplies a legal finite-memory control for every finite-rate state. Under that control the ergodic theorem and the terminal-prefix disintegration make the stopped state converge to the target, and the normalized entropy converges to its rate. This proves the upper bound in the variational representation and hence the Laplace principle.

For physical time, finite horizon supplies a continuous strictly increasing clock on every rate sublevel. Proposition 1.3 gives continuity of the generalized inverse away from the retained terminal atom. The terminal prefix resolves precisely that atom, so the extended contraction principle applies. The recovery controls may alter their final complete excursion by a connector

of $o(T)$ cost to hit the physical horizon; the prefix entropy formula accounts for the remaining fractional excursion. $\square$

Let C_N be a joint homology–return–roof conditioning window in one of the A2 regimes. For a bounded continuous stopped functional F , the inserted A2 Fourier theorem applies to $\mathbb{E}[e^{-NF}\mathbf{1}_{C_N}]$ after the usual exponential tilting of the finite-memory approximants.

Theorem 1.8 (Conditional stopped-path LDP). *For every admissible A2 local, central, or saturated window whose limiting constraint lies in a compact interior saddle chart, the conditional stopped states satisfy an LDP. Its rate is*

$$\mathcal{I}_c^C(z) = \mathcal{I}_c(z) - \inf\{\mathcal{I}_c(y) : y \in C\}$$

on the exact constraint set and is $+\infty$ off it; the analogous statement holds in physical time.

Proof. The denominator has the positive sharp asymptotic of A2. For the numerator, apply the same inserted Fourier decomposition after the finite-memory recovery reduction. The insertion is a bounded multiplier on the A2 bundle, and approximation errors are exponentially negligible by Lemma 1.5. Thus numerator and denominator have the same local lattice–continuous prefactor at each constrained saddle; their exponential parts differ by the infimum of $\mathcal{I}_c + F$ and that of $\mathcal{I}_c$. Taking logarithms gives the conditional Laplace principle. Goodness follows by intersecting a good-rate sublevel with the closed constraint set. $\square$

Theorem 1.9 (Round-twenty-one A3 closure). *The Sinai empirical-path theory is built on the exact continuous marked-history kernel, a Polish stopped/recession state, entropy-controlled compactness, a legal recovery theorem, explicit good rates, and sharp conditional Laplace principles. No one-step countable surrogate and no point-probability formula for a continuous prefix is used.*

Proof. Combine Theorems 1.1, 1.4, 1.6, 1.7, and 1.8. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states a positive theorem reconstruction in the model-specific regular regime declared in its controlling theorems. Cross-paper inputs follow the Round-Twenty-One dependency ledger. Repository source identity, proof-structure, counterexample, and build checks are internal reproducibility gates and do not replace independent external mathematical review.

PEKING UNIVERSITY, BEIJING 100871, CHINA
Email address: qiqian@pku.edu.cn